Artificial Intelligence and Search

A language model is a neural network trained to predict and generate sequences of text. Tokenization divides text into smaller units called tokens so a model can process words and punctuation. An embedding is a numeric vector that represents the meaning of a text passage; passages with similar meanings tend to have nearby vectors. A vector database stores embeddings and can quickly find passages whose vectors are close to a query vector. Retrieval augmented generation (RAG) first retrieves relevant passages from a collection and then gives those passages to a language model as evidence for its answer. This design can ground answers in a private document collection and attach source citations.